MICHAEL KASMAN

Data Scientist

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EDUCATION

B.S. in Computer Engineering
Junior - Class of May 2022

The University of Texas at Dallas *Richardson, TX*

Calculus I, II, III • Differential Equations • Linear Algebra • Probability & Statistics • Computer Science I, II • Data Structures & Algorithms

WORK EXPERIENCE

Undergraduate Researcher - Hand Synergies
The University of Texas at Dallas

June 2020 - August 2020 *Richardson, TX*

Achievements/Tasks

- Investigated non-linear dimensionality reduction techniques on hand joint angle measurements for real-time application under Dr. Ann Majewicz Fey & Dr. Edoardo Battaglia in the HERO Lab
- Prepared hand joints dataset to train and test dimensionality reduction models: **principal component analysis (PCA), kernel PCA, UMAP,** and **autoencoder**
- Utilized **Keras, TensorFlow,** and **Sklearn** to build a **stacked autoencoder** with a reconstruction error of **≈ 3.6%** (less than PCA)

Undergraduate Researcher - Robotic Surgery
The University of Texas at Dallas

May 2019 - August 2019 *Richardson, TX*

Achievements/Tasks

- Researched the variance of human-centric metrics between virtual reality simulation and physical dry-lab exercises in robotic surgical training under Dr. Ann Majewicz Fey in the HERO Lab
- Pre-processed and analyzed data in a Linux environment using **Pandas**, **NumPy,** and **Seaborn** in **Jupyter** Lab to visualize analysis of variance **(ANOVA)** and **post-hoc Tukey comparison**
- Published work to the Journal of Medical Robotics Research for special issue: Technology-enabled Tools for Clinical Skills Assessment

Clark Foundation Summer Research Scholar
The University of Texas at Dallas

June 2018 - August 2018 *Richardson, TX*

Achievements/Tasks

- Developed a mobile indoors navigation app, Constellation, for the UT Dallas campus by collaborating with undergraduate researchers and principal investigator, Dr. Ravi Prakash
- Created **PHP** scripts to receive HTTP requests from the client and acquire the requested path coordinates and user location from **ArcGIS** and **Cisco CMX**
- Accomplished Constellation 1.0, measured by 50+ successful consecutive tests in navigating a path between rooms, through frontend and backend communication

PUBLICATION

Michael Kasman, Ziheng Wang, Marco Martinez, Robert Rege, Herbert Zeh, Daniel Scott, and Ann Majewicz Fey. A Comparative Human-Centric Analysis of Virtual Reality and Dry Lab Training Tasks on the da Vinci Surgical Platform. *Journal of Medical Robotics Research: Special Issue on Technology-enabled Tools for Clinical Skills Assessment.*

SKILLS

Bootstrap | SQL | LaTex | Flask
Linux | NumPy | Matplotlib | Seaborn
Pandas | Sklearn | Keras | TensorFlow

HTML 5/CSS 3	**Proficient**
Python	**Proficient**
JavaScript	**Intermediate**
Java	**Intermediate**
C/C++	**Intermediate**
PHP	**Intermediate**

PROJECTS

Flare

- A **React Native** mobile app that utilizes a **convolutional neural network** to classify the victim density level of natural disasters from aerial/drone-eye view images
- Integrated **TensorFlow, Keras, Sklearn,** and **Jupyter** to construct and optimize the neural network

Kaggle Medical Cost Predictor

- A **Python** Notebook analyzing health insurance data and predicting medical charges from age, sex, BMI, smoking status, and number of dependents
- Built a **multiple linear regression** model and performed data analysis with **Sklearn, NumPy, Pandas, Matplotlib,** and **Seaborn**

Smart Rates

- A responsive web-app that graphs the real-time market suggested retail price of cars using a **Flask** server integrated with the Smart Car API

Constellation Mobile Indoors Navigation App

- An **Android** mobile indoors navigation app capable of location tracking and path-finding between rooms on the UT Dallas campus

GreenView

- An **Android** mobile app that determines whether an item is recyclable through image recognition with the Clarifai API and **MySQL** database of items

ACHIEVEMENTS

2020 Jonsson School Undergraduate Research Award
Awarded $500, among five other recipients, for medical robotics research at the University of Texas at Dallas

IEEE-Eta Kappa Nu (IEEE-HKN)
Recognized in the top quarter of class in Computer Engineering at the University of Texas at Dallas

2017 cPanel & FreeCodeCamp Hackathon 1st Place
Won 1st Place with three team members out of 50+ teams for the best mobile app, GreenView, that identifies the recyclability of items